

Prasad's Notes for Cyber Forensics

CYBER FORENSICS

Complete Study Notes & 40-Question MCQ Practice Set

Simple English • Diagrams for Quick Revision • Important Points Highlighted

How to Use These Notes

- Read each Unit in order — later units (hashing, imaging, tools) build on earlier ones (file systems, evidence handling).
- Boxes marked ★ IMPORTANT are exam-favourite points — memorise these first.
- Diagrams are for quick revision the night before the exam — if you can redraw a diagram from memory, you know the topic.
- Attempt the 40 MCQs at the end without looking at answers first, then check the explanations.

Unit 1: Introduction to Computer Forensics

1.1 What is Computer Forensics?

Computer Forensics (also called Digital Forensics) is the science of finding, preserving, analysing, and presenting digital evidence in a way that is acceptable in a court of law. In simple words — it is like being a detective, but instead of searching a physical crime scene, you search computers, phones, and networks for clues.

The evidence could be a deleted file, an email, a log entry, a browser history, or even a single timestamp. The forensic investigator's job is to find this evidence WITHOUT damaging or changing it, because changed evidence is often useless in court.

★ **IMPORTANT:** Computer forensics is not just about 'finding' evidence — it is about finding it in a way that keeps it legally valid (unchanged, well documented, and explainable).

1.2 Computer Crime vs Unauthorized Activities

These two terms sound similar but are different in scope:

Aspect	Computer Crime	Unauthorized Activity
Meaning	An act that breaks a specific law (punishable by courts)	Any action done without permission — may or may not break a law
Example	Hacking a bank server and stealing money	An employee using office Wi-Fi for personal browsing
Legal action	Criminal case can be filed	Usually only internal/company action (warning, termination)
Severity	Generally serious, well-defined offence	Can range from minor policy breach to serious misuse

In short: every computer crime is an unauthorized activity, but not every unauthorized activity is a computer crime.

1.3 Cyber Laws

Cyber Laws are the legal rules that govern activities on the internet and with computers — covering data protection, hacking, cyber fraud, digital signatures, and privacy. In India, the main law is the Information Technology (IT) Act, 2000, which was later amended in 2008 to cover new offences like cyber terrorism and data breaches.

- Cyber laws define what counts as a cybercrime (hacking, phishing, identity theft, cyberstalking, etc.).
- They give legal recognition to digital signatures and electronic records.
- They define the punishment and the procedure to follow when a cybercrime happens.
- Forensic evidence collected must follow these laws to be admissible (acceptable) in court.

1.4 The Six-Step Process of Computer Forensics

Every forensic investigation — big or small — follows the same six steps. This is one of the most important things to remember for exams.

The Six-Step Computer Forensics Process



Figure 1.1 — The Six-Step Computer Forensics Process

Step	What Happens
1. Identification	Recognise that an incident has happened and identify what evidence (devices, files, logs) might be relevant.
2. Preservation	Protect the evidence from being changed, damaged, or deleted — e.g., by write-blocking a hard disk.
3. Collection	Actually gather the evidence — imaging disks, copying logs, taking photos of the scene.
4. Examination	Use forensic tools to uncover hidden, deleted, or encrypted data from the collected evidence.
5. Analysis	Interpret the data found — connect the dots to understand what happened, when, and by whom.
6. Reporting	Document all findings clearly so they can be understood by non-technical people (lawyers, judges, managers).
★ IMPORTANT: Remember the order using the short-form: I-P-C-E-A-R (Identify, Preserve, Collect, Examine, Analyse, Report).	

1.5 Why Do We Need a Forensic Investigator?

- Ordinary IT staff may accidentally destroy evidence while trying to 'fix' a hacked system.
- A trained investigator knows how to collect evidence without changing timestamps or file content.
- They understand the legal requirements so evidence can be used in court.
- They can uncover hidden/deleted data using specialised tools and techniques.
- They provide expert testimony in court, explaining technical findings in simple terms.

Unit 2: File Systems & Data Storage Basics

2.1 What is a File System?

A file system is the method an operating system uses to store, organise, and retrieve files on a storage device (hard disk, USB, SD card). Think of it like a library's cataloguing system — without it, the OS would not know where each file physically lives on the disk.

Common file systems: FAT32, exFAT, NTFS (Windows), ext4 (Linux), APFS (Mac).

2.2 Logical vs Physical Storage

Aspect	Logical Storage	Physical Storage
Meaning	How the OS/file system 'sees' and organises data (drives, folders, files)	How data is actually stored on the physical hardware (platters, sectors, memory chips)
Example	C:\Users\John\file.docx	Data written on specific sectors/blocks of the disk
Forensic relevance	Logical view may hide deleted or fragmented data	Physical (bit-by-bit) view captures everything, including hidden/deleted data

2.3 Metadata & Timestamps (MAC Times)

Metadata is 'data about data' — information describing a file without being the file's actual content, such as size, owner, and timestamps. In forensics, the three most important timestamps are remembered using the short form MAC:

Timestamp	Meaning
Modified (M)	When the file's CONTENT was last changed.
Accessed (A)	When the file was last opened/read (even without changes).
Created (C)	When the file was first created on that particular storage location.
★ IMPORTANT: MAC times can be manipulated (timestomping) by attackers to hide their tracks — e.g., changing a malicious file's 'Created' date to look older. Investigators cross-check MAC times with other logs to catch this.	

2.4 FAT32 vs NTFS

FAT32 keeps track of files using a File Allocation Table (FAT) — a simple index listing which disk clusters belong to which file. NTFS (New Technology File System) instead uses a Master File Table (MFT), a much more detailed database where every file has its own record containing metadata, permissions, and even small file content.

FAT32 vs NTFS — Quick Comparison

FAT32	NTFS
Uses File Allocation Table (FAT)	Uses Master File Table (MFT)
Max file size: 4GB	Max file size: very large (TBs)
No built-in security (permissions)	Has permissions, encryption (EFS)
Weaker recovery of deleted files	Journaling helps recovery & forensics
Simple, older systems	Modern Windows default

Figure 2.1 — FAT32 vs NTFS

- **File linking:** In FAT32, a file's clusters are found by following a chain in the FAT. In NTFS, each file has an MFT entry that directly points to its data (faster, more reliable).
- **Deleted file recovery:** In both systems, deleting a file usually only removes its FAT/MFT entry, not the actual data — this is why recovery tools can 'undelete' files.

2.5 What Really Happens When You Delete a File

What Really Happens When a File Is Deleted



Figure 2.2 — File Deletion Concept

A very common misconception is that 'delete' = 'erased forever'. In reality, deleting a file simply tells the file system 'this space is now free to use' — the actual 1s and 0s remain on the disk until the operating system overwrites that space with new data.

★ **IMPORTANT:** File deletion ≠ data erased. This single fact is the foundation of most data-recovery and forensic-investigation techniques.

2.6 Slack Space & Unallocated Space

Term	Meaning
Slack Space	The unused leftover space at the end of the last cluster allocated to a file (because file sizes rarely fit clusters)

Term	Meaning
	exactly). Old data from a previously deleted file can remain here.
Unallocated Space	Disk space marked as 'free' by the file system, not currently linked to any file. Deleted files often still exist here until overwritten.

2.7 File Signatures (Magic Numbers)

Every file type has a unique 'signature' — a few bytes at the very beginning of the file that identify its true format, regardless of what extension it has been given. Investigators use this to catch criminals who rename files (e.g., renaming a .jpg to .txt to hide it).

File Type	Signature (Hex)	Notes
JPG	FF D8 FF	Found at the start of every JPEG image
PNG	89 50 4E 47	Spells 'PNG' in ASCII after the first byte
PDF	25 50 44 46 (%PDF)	Literally starts with the text %PDF
DOCX/ZIP	50 4B 03 04 (PK..)	DOCX/XLSX/PPTX are actually ZIP archives
EXE	4D 5A (MZ)	Stands for 'Mark Zbikowski', the DOS executable designer

★ **IMPORTANT:** A file's extension can be changed by anyone, but its signature (magic number) cannot lie — forensic tools always verify the actual file type using the signature, not the extension.

Unit 3: Computer Forensics Techniques & Procedures

3.1 What Computer Forensics Involves

- Preservation — keeping evidence safe and unchanged.
- Identification — recognising what is relevant evidence.
- Extraction — pulling the actual data/evidence out from the device.
- Documentation — recording every action taken, with timestamps and names.
- Interpretation — explaining what the evidence means in the context of the case.

3.2 Goals of Forensic Analysis

- To reconstruct exactly what happened during an incident (the 'story' of the crime).
- To identify who was responsible.
- To find out how the attacker got in (the vulnerability or method used).
- To gather evidence strong enough to stand up in a court of law.

3.3 Types of Cyber Forensics Techniques

Technique	Focus Area
Disk Forensics	Examining hard disks/SSDs for files, deleted data, and metadata
Network Forensics	Monitoring & analysing network traffic to detect intrusions
Memory (RAM) Forensics	Analysing volatile memory for running processes, passwords, malware
Mobile Forensics	Extracting data (calls, messages, apps) from smartphones/tablets
Email Forensics	Tracing email headers, senders, and attachments for fraud/phishing cases
Cloud Forensics	Investigating data stored on cloud servers (AWS, Azure, etc.)

3.4 Cyber Forensics Procedure: Preparation

What to do BEFORE an incident happens

- Set up logging and monitoring systems in advance.
- Prepare an Incident Response Plan (IRP) — a documented, step-by-step guide for handling incidents.
- Form and train an Incident Response Team (IRT) with clearly defined roles.
- Keep forensic toolkits (write blockers, imaging software) ready and tested.

3.5 Incident Response Life Cycle

Incident Response Life Cycle

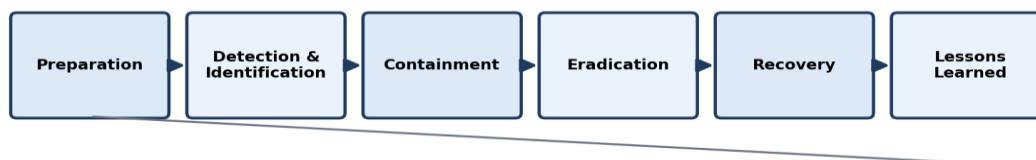


Figure 3.1 — Incident Response Life Cycle

This cycle repeats — 'Lessons Learned' feeds back into 'Preparation', so the organisation keeps improving after every incident.

3.6 Detecting Incidents

Incidents are usually detected through: unusual log entries, alerts from security tools (IDS/IPS, antivirus), unexpected system slowdowns, unfamiliar processes, or reports from users/customers noticing something wrong.

3.7 Chain of Custody

Chain of Custody is the chronological, documented record showing who collected, handled, transferred, and stored a piece of evidence, and when. If there is a gap or unexplained access in this chain, the evidence can be challenged and rejected in court.

Chain of Custody



Chain of Custody = an unbroken, documented trail proving evidence was never tampered with.

Figure 3.2 — Chain of Custody

★ **IMPORTANT:** A broken chain of custody is one of the most common reasons digital evidence gets thrown out of court — always document every single handoff.

Unit 4: Networking, Processes & Logs for Forensics

4.1 OS vs Application

Aspect	Operating System (OS)	Application
Role	Manages hardware & provides base services (Windows, Linux)	Runs on top of the OS to perform a specific task (browser, Word)
Example	Windows 11, Ubuntu Linux	Chrome, MS Word, WhatsApp
Forensic relevance	OS logs show system-wide events (logins, boot)	Application logs show what a specific program did (browsing history, chat logs)

4.2 User Mode vs Kernel Mode

- User Mode: Where normal applications run, with limited/restricted access to hardware — if an app crashes here, the whole system usually survives.
- Kernel Mode: Where the OS core and device drivers run, with full access to hardware and memory — a crash here can bring down the whole system. Malware often tries to get into kernel mode for maximum control (rootkits).

4.3 Processes, Process States, PID & PPID

A process is a program currently being executed. Every process is given a unique Process ID (PID). The process that created it is called the Parent Process, identified by its Parent Process ID (PPID).

- Process states typically include: New, Ready, Running, Waiting, and Terminated.
- Forensic value: Malware often spawns unusual parent-child relationships (e.g., Microsoft Word suddenly launching PowerShell) — this is a major red flag investigators look for.

★ **IMPORTANT:** Why malware uses threads: Threads let a single malicious process perform several tasks at once (e.g., stealing data while also hiding itself) using fewer resources and making detection harder.

4.4 Network Traffic Analysis

Concept	Meaning
Packet Capture	Recording raw network traffic as it happens (using tools like Wireshark/tcpdump).
Packet Analysis	Studying already-captured packets to understand communication, detect attacks, etc.
Live Capture	Capturing and viewing traffic in real time, directly from the network interface.
PCAP Analysis	Analysing a saved .pcap file (recorded traffic) after the fact — no time pressure.
Promiscuous Mode	A network card setting that allows it to capture ALL traffic on the network segment, not just packets addressed to it.

A Network Packet — What It Contains

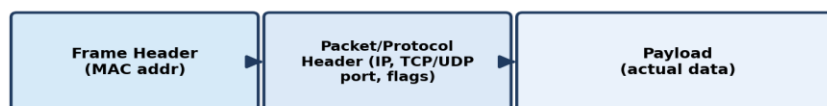


Figure 4.1 — Structure of a Network Packet

Capture Filters vs Display Filters

- Capture Filters: Applied BEFORE capturing — they decide what traffic actually gets recorded (reduces file size).
- Display Filters: Applied AFTER capturing — they only change what you SEE on screen; all captured data is still stored.

Common Wireshark Filters

Filter	Purpose
ip.addr == x.x.x.x	Show traffic to/from a specific IP address
tcp.port == 80	Show traffic on a specific TCP port
udp.port == 53	Show traffic on a specific UDP port (e.g., DNS)
http	Show only HTTP traffic
dns	Show only DNS query/response traffic
icmp	Show ping/ICMP traffic
tcp.flags	Filter based on TCP flags (SYN, ACK, FIN, RST) — useful for spotting scans

4.5 Logs — The Digital Diary of a System

- Event: Something that actually happens on a system (e.g., a user logs in).
- Log Entry: The recorded, written trace of that event, stored by the OS/application.
- Logs are usually written to storage in near real-time by the OS logging service, then rotated/archived after reaching a size or time limit.

Local Time vs UTC & Epoch Time

- Local Time: The time as per the device's configured time zone — can vary by device/location.
- UTC (Coordinated Universal Time): A single global time standard, unaffected by time zones — investigators always convert logs to UTC to compare evidence from different locations correctly.
- Epoch Time: The number of seconds elapsed since 1 January 1970 (UTC) — a universal numeric way computers store time internally.

★ **IMPORTANT:** Never trust a raw timestamp until you know its time zone. Mixing local time and UTC logs is a classic forensic mistake that can completely change the timeline of events.

4.6 Common Windows Event IDs

Event Type	What It Tells the Investigator
Logon/Logoff	Who accessed the system, and when — including failed login attempts (brute force clues)
Process Creation	Which programs were launched, by whom, with what command-line arguments
Account Changes	New accounts created, passwords reset, group memberships changed
Privilege Escalation	A user/process gained higher permissions — a strong sign of an attack in progress

Key Linux Log Files

- /var/log — main directory storing most Linux log files.
- auth.log — records authentication attempts (SSH logins, sudo usage).
- syslog — general system activity log.
- bash_history — records commands typed by a user in the terminal (very useful, but attackers often clear this).

Unit 5: Evidence Handling & Investigation

5.1 Evidence Checkout Log

A formal register that records every time evidence is taken out of storage and returned — including the name of the person, date/time, purpose, and signature. This log is a key part of proving chain of custody.

5.2 Handling Evidence & First Response

- First Response: The immediate actions taken by the first person who arrives at the scene of a digital incident — the goal is to preserve evidence, NOT to start investigating immediately.
- Never turn off a running computer without proper procedure (volatile data in RAM could be lost) — unless following a specific 'pull the plug' policy for certain cases.
- Photograph the scene, note what is on the screen, and label all cables/devices before touching anything.

5.3 Formulating & Executing a Response Strategy

Before jumping into action, the team formulates a response strategy based on the type of incident, the systems affected, and business priorities. This strategy is then executed carefully and documented step by step.

5.4 Forensic Duplication

This means creating an exact, verified copy (image) of the original evidence so that ALL further analysis is done on the copy — the original is preserved untouched, exactly as first collected.

5.5 Authenticating the Evidence

Authentication proves that the evidence collected is exactly the same as what existed at the crime scene, with no changes. This is usually proved using hash values (see Unit 7) — if the hash of the evidence matches the hash taken at collection time, the evidence is authentic.

★ **IMPORTANT:** Golden Rule of Forensics: Never work on the original evidence. Always create a verified image/copy first, and perform all analysis on that copy.

5.6 Investigation, Common Mistakes & Detection

Common Mistakes Investigators Must Avoid

- Working directly on original evidence instead of a copy.
- Not documenting each step (breaks chain of custody).
- Shutting down a live system incorrectly, losing volatile (RAM) data.
- Using un-validated/unlicensed tools that a court may not accept.
- Ignoring a hash mismatch instead of investigating why it happened.

5.7 The Initial Assessment & Incident Notification Checklist

The Initial Assessment is a quick, early evaluation to understand the scope and severity of an incident (what systems, how many users, how critical). Based on this, an Incident Notification Checklist is followed to make sure the right people (management, legal, law enforcement, affected customers) are informed at the right time, in the right order.

Unit 6: Hexadecimal Notation

6.1 What is Hexadecimal?

Hexadecimal (base-16) is a number system using 16 symbols: 0-9 and A-F (where A=10, B=11 ... F=15). Computers store everything in binary (base-2), but binary is very long and hard for humans to read — hexadecimal packs the same information into a much shorter, more human-friendly form (each hex digit = exactly 4 binary bits).

Hexadecimal — Quick Reference Table

Hex	Binary	Decimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
A	1010	10
B	1011	11
C	1100	12
D	1101	13
E	1110	14
F	1111	15

Figure 6.1 — Hex / Binary / Decimal Quick Reference

6.2 Why is Hex Used in Forensics?

- Raw disk/file data is shown in Hex Editors so investigators can view the exact bytes of a file, including hidden/deleted content.
- File signatures (magic numbers) are written and checked in hex, e.g., FF D8 FF for JPG.
- Memory addresses, MAC addresses, and hash values are commonly represented in hexadecimal.

★ **IMPORTANT:** Two hex digits = exactly one byte. This is why hex is the 'natural language' for looking at raw computer data.

Unit 7: Hashing, Windows Registry & System Artefacts

7.1 Hashing Before Imaging vs After Imaging

A hash is a fixed-length 'digital fingerprint' calculated from data using an algorithm (MD5, SHA-1, SHA-256). Before creating a forensic image, the investigator calculates the hash of the original evidence. After imaging, the hash of the new image (the destination hash) is calculated again.

Term	Meaning
Hash Before Imaging	The 'fingerprint' of the ORIGINAL evidence, taken before any copying begins.
Destination Hash	The fingerprint of the newly created IMAGE/copy, taken right after imaging.
Hash Comparison	Comparing the two hashes — if they MATCH, the image is a perfect, unaltered copy.
★ IMPORTANT: If a hash mismatch happens and is IGNORED, the entire forensic image becomes legally worthless — a mismatch means the copy is NOT identical to the original, and this must always be investigated, never skipped.	

7.2 Windows Registry — A Goldmine of Evidence

The Windows Registry is a hierarchical database storing configuration settings for the OS and installed applications. For forensic investigators, it is an extremely rich source of evidence — it can show installed software, USB devices ever connected, recently opened files, and user activity.

Windows Registry — Key Hives (Forensic Gold Mine)

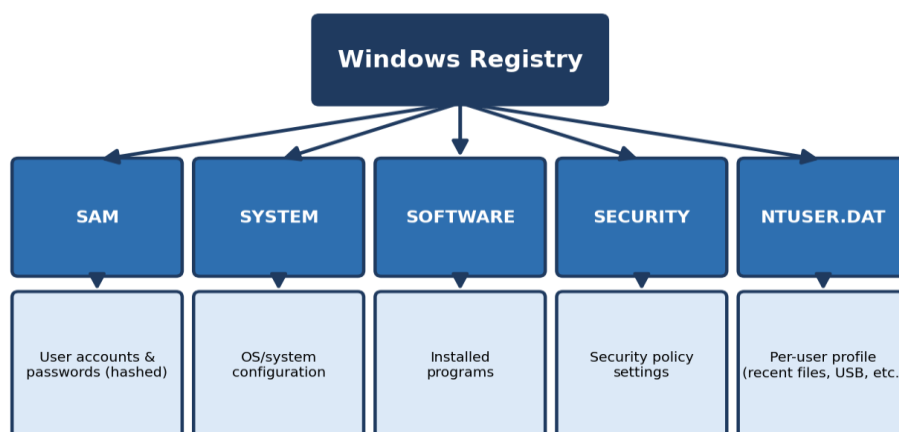


Figure 7.1 — Key Windows Registry Hives

Hive	What It Stores
SAM (Security Account Manager)	Local user account names and hashed passwords

Hive	What It Stores
SYSTEM	Hardware and driver configuration, boot settings
SOFTWARE	Installed applications and their settings
SECURITY	Local security policy and audit settings
NTUSER.DAT	Per-user settings — recent documents, USB history, desktop preferences

7.3 Linux System Artefacts

Command / File	Purpose
ps	Lists currently running processes
top	Shows live, real-time resource usage per process
/proc	A virtual filesystem exposing live kernel & process information
/var/log	Main directory for Linux system log files
auth.log	Authentication attempts (SSH, sudo)
syslog	General system messages/events
bash_history	History of commands typed by a user

7.4 BIOS/UEFI & Boot Sequence

BIOS (Basic Input/Output System) and its modern replacement UEFI (Unified Extensible Firmware Interface) are the first software that runs when a computer is powered on. They perform hardware checks (POST) and then hand control over to the Bootloader, which locates and loads the operating system into memory.

1. Power On → BIOS/UEFI performs POST (Power-On Self-Test).
2. BIOS/UEFI locates the boot device (hard disk, USB, network).
3. Bootloader (e.g., GRUB, Windows Boot Manager) is loaded.
4. Bootloader loads the Operating System kernel into memory.
5. OS takes over and finishes startup (loading drivers, services, login screen).

Unit 8: Encoding, Encryption & Hashing

8.1 Encoding vs Encryption vs Hashing

Students very commonly confuse these three — but they serve completely different purposes:

Encoding vs Encryption vs Hashing

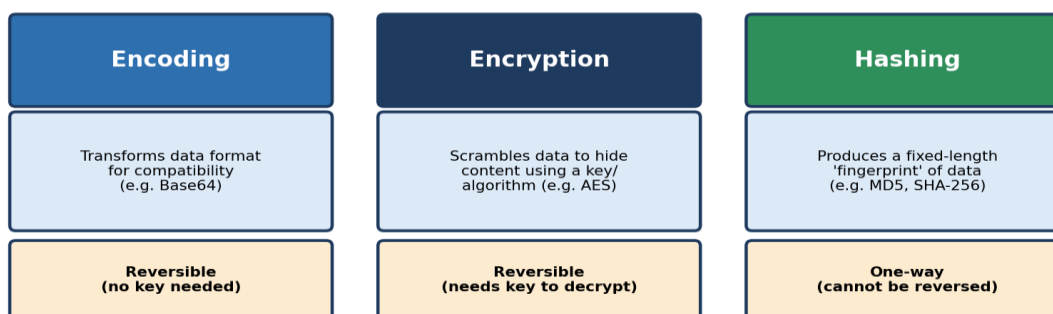


Figure 8.1 — Encoding vs Encryption vs Hashing

Feature	Encoding	Encryption	Hashing
Purpose	Format compatibility (e.g., Base64 for email)	Confidentiality — hide data from unauthorised viewers	Integrity — verify data hasn't changed
Reversible?	Yes, no key needed	Yes, but only with the correct key	No — one-way, cannot get original back
Example	Base64, URL encoding	AES, RSA	MD5, SHA-256

8.2 The Hex Editor

A Hex Editor is a tool that shows the raw bytes of a file in hexadecimal (and often the equivalent ASCII text alongside). Investigators use it to inspect file headers, find hidden data, verify file signatures, and study data that normal applications cannot open or interpret correctly.

8.3 Hashing Algorithms & Hash Collisions

A good hash function should produce a completely different output for even a tiny change in input, and no two different inputs should ever produce the same hash. When two different inputs DO produce the same hash, it is called a hash collision.

- MD5 Hash Collisions: MD5 (Message Digest 5) is now considered 'broken' for security purposes because researchers have shown it is possible to deliberately create two different files with the same MD5 hash.
- Because of this weakness, modern forensic and security practice prefers SHA-256 (or stronger) over MD5 wherever strong collision-resistance is required.

★ **IMPORTANT:** A hash collision does not mean the data is identical — it just means two different pieces of data accidentally (or deliberately) produced the same fingerprint. This is why MD5 alone is no longer trusted for high-security integrity checks.

8.4 Bit Rot

Bit Rot (also called data degradation) is the slow, natural corruption of data stored on physical media over time, caused by physical decay of the storage medium, magnetic field weakening, or cosmic radiation. Even without anyone touching the file, a bit can silently flip from 0 to 1 or vice versa, which is another reason hash verification is done regularly on long-term evidence storage.

Unit 9: Standard Operating Procedures & Crime Scene Processing

9.1 Standard Operating Procedures (SOPs)

SOPs are formally documented, step-by-step instructions that ensure every investigator handles evidence and incidents in the SAME consistent, repeatable way — regardless of who is doing the work. Following SOPs is essential to make forensic findings defensible in court.

- Reduces human error and inconsistency between investigators.
- Provides a documented, defensible process if evidence is challenged in court.
- Makes training new investigators easier and faster.

9.2 Processing Crime & Incident Scenes — Windows & DOS Systems

When processing a Windows/DOS system as a crime scene, the investigator follows a careful, non-destructive sequence:

6. Secure the scene — restrict physical access to the device.
7. Document the current state (screen contents, running programs, cable connections) with photos.
8. Decide whether to keep the system running (for volatile RAM evidence) or shut it down (following SOP).
9. Use a write blocker before connecting any storage device to the forensic workstation.
10. Create a forensic image and calculate hash values immediately.
11. Package, label, and log the original evidence, then begin analysis only on the image copy.

Unit 10: Forensics Implications, Standards & Privacy

10.1 Forensic Implications

Every action a forensic investigator takes can have legal, technical, and organisational consequences ('implications'). For example, improperly seizing a device can violate someone's rights; a badly documented investigation can get a whole case dismissed.

10.2 Accreditation Standards

Accreditation Standards (such as ISO/IEC 17025 for forensic laboratories) are official benchmarks that a forensic lab or process must meet to be considered trustworthy and competent. Courts give more weight to evidence coming from an accredited lab following recognised standards.

10.3 Performing a Cyber Forensics Investigation

A complete investigation combines everything learnt so far: identifying evidence, preserving it, following chain of custody, imaging with hash verification, analysing with proper tools, and reporting findings clearly and objectively — without bias toward any particular conclusion.

10.4 Privacy and Cyber Forensics

Investigators must always balance the need to collect evidence with an individual's right to privacy. This means only collecting data that is relevant to the case (not everything on a device), following legal warrants/authorisation, and protecting sensitive personal data (like someone's medical records or private photos) that is not related to the investigation.

★ **IMPORTANT:** An investigation that violates privacy laws while collecting evidence can result in that evidence being declared inadmissible — legal, ethical process matters as much as technical skill.

Unit 11: Forensic Imaging

11.1 Why Imaging Instead of Working on the Original?

If you make even one small change while investigating the original evidence (opening a file, running a program), you can permanently alter timestamps or overwrite deleted data — destroying evidence. Working on a verified image instead means the original stays untouched and can always be re-checked.

11.2 Copying vs Forensic Imaging

Aspect	Normal Copying	Forensic Imaging
What is copied	Only visible files and folders	EVERY bit on the disk — including deleted files, slack space, unallocated space
Metadata preserved?	Often changes timestamps	Preserves exact original metadata
Verified with hash?	No	Yes — always hashed before and after
Court acceptable?	Generally not sufficient alone	Yes, when done correctly with documentation

11.3 Types of Imaging

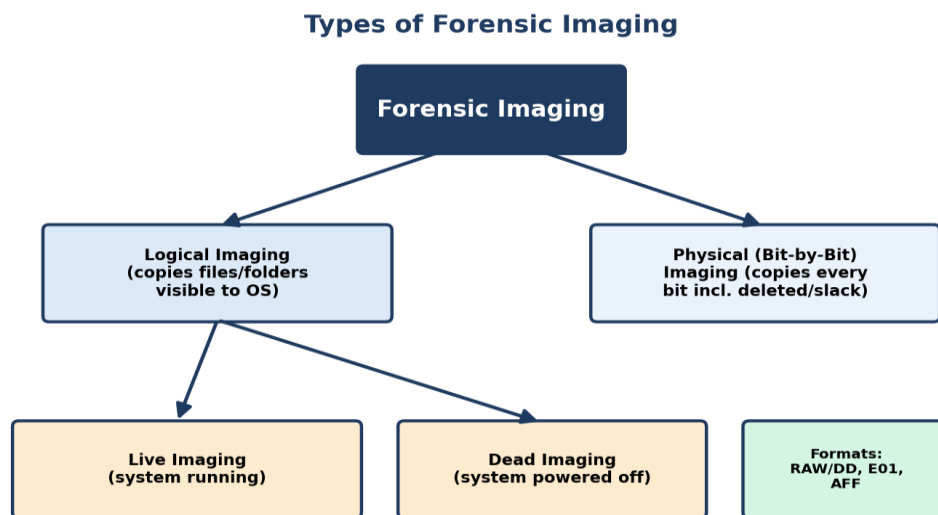


Figure 11.1 — Types of Forensic Imaging

Type	Meaning
Logical Imaging	Copies only the files/folders visible to the operating system (not deleted/hidden data)
Physical (Bit-by-Bit) Imaging	Copies every single bit of the storage device, including deleted files and slack space — the gold standard for forensics
Live Imaging	Imaging performed while the system is still running (used when shutting down would lose important volatile data)

Type	Meaning
Dead Imaging	Imaging performed after the system has been powered off — safer, but loses RAM/volatile data

11.4 Common Forensic Image Formats

Format	Full Form / Notes
RAW (DD)	A plain bit-for-bit copy with no extra metadata; created using the classic 'dd' command; universally compatible
E01 (EnCase Evidence File)	Compressed format with built-in metadata and hash values, created by EnCase/FTK tools
AFF (Advanced Forensic Format)	An open-source image format designed specifically for forensic use with built-in integrity checks

11.5 Write Blockers

A Write Blocker is a hardware device or software tool that allows an investigator to READ data from a storage device while physically or logically PREVENTING any data from being written to it. This guarantees the original evidence cannot be accidentally modified during imaging.

Type	How It Works
Hardware Write Blocker	A physical device connected between the evidence drive and the forensic workstation; intercepts and blocks all write commands at the hardware level
Software Write Blocker	A program running on the forensic workstation's OS that blocks write commands at the software/driver level
★ IMPORTANT: Imaging without a write blocker is one of the most serious mistakes in forensics — even simply connecting a drive to a normal computer can trigger automatic writes (like updating a 'last accessed' timestamp), contaminating the evidence and making it vulnerable to rejection in court.	

Also Remember

- File-level imaging (imaging just a single file) is acceptable only when the whole-disk image is not necessary or not possible — e.g., copying just one suspicious document from a live production server.
- Imaging errors/risks include: incomplete image (bad sectors skipped), hash mismatch, and evidence contamination — courts view any unexplained contamination as a serious threat to the evidence's credibility.

Unit 12: Computer Forensics Tools

Tool	What It Does
Sysinternals Suite	A collection of free Microsoft utilities (Process Explorer, Autoruns, TCPView, etc.) for deep Windows system analysis
FTK (Forensic Toolkit)	A complete forensic analysis suite used to index, search, and analyse evidence (emails, files, registry) at scale
FTK Imager	A free tool specifically for creating forensic images and previewing evidence without altering it
OSForensics (OSF)	An all-in-one forensic tool for indexing files, recovering deleted data, and analysing memory/registry
Hex Editors	Tools (e.g., HxD, WinHex) for viewing/editing the raw hexadecimal content of files and disks

★ **IMPORTANT:** FTK Imager is used to **CREATE** and **PREVIEW** evidence images (does not do deep analysis); full FTK is used to **ANALYSE** that evidence in depth. Don't confuse the two in exams!

Unit 13: Live System, Linux & Mobile Forensics

13.1 Live System Forensics

Live forensics means collecting and analysing evidence from a system that is still running (powered on), rather than shutting it down first. This is essential for capturing volatile data — information that would be lost the instant the power is cut, such as: running processes, open network connections, contents of RAM, and logged-in user sessions.

★ **IMPORTANT:** Order of Volatility — always collect the MOST volatile data first: RAM/memory → network connections → running processes → disk data → archived logs/backups (least volatile, safest to collect last).

13.2 Linux Forensics

Collecting and analysing artefacts from Linux systems relies heavily on its transparent, text-based logging and configuration. Key areas: /var/log directory, auth.log/syslog, bash_history, running processes (ps, top), and the /proc virtual filesystem for live kernel/process data.

13.3 Introduction to Mobile Forensics

Mobile Forensics is the branch of digital forensics focused on recovering evidence from smartphones and tablets — call logs, SMS/messages, app data, GPS location history, and photos. It presents unique challenges compared to computer forensics:

- Wide variety of devices, operating systems (Android/iOS), and manufacturers, each with different security models.
- Data can be remotely wiped if the device stays connected to a network — investigators use Faraday bags to block signals during seizure.
- Encryption and lock screens can make direct access very difficult without specialised tools or legal cooperation from the manufacturer.

Quick Revision — One-Page Summary

Concept	One-Line Memory Hook
Forensics Process	Identify → Preserve → Collect → Examine → Analyse → Report
File Deletion	Deleting only removes the pointer, not the data (until overwritten)
MAC Times	Modified, Accessed, Created — can be manipulated (timestamping)
FAT32 vs NTFS	FAT = File Allocation Table (older); NTFS = MFT (modern, secure)
Chain of Custody	Unbroken documented trail — a gap can get evidence rejected in court
Hashing	One-way fingerprint; used to prove evidence is unchanged
Hash Mismatch	Never ignore — always investigate the cause
Write Blocker	Allows reading but blocks writing — protects original evidence
Imaging Types	Logical (files only) vs Physical (every bit, incl. deleted data)
Encoding vs Encryption vs Hashing	Encoding = format; Encryption = secrecy (reversible); Hashing = integrity (one-way)
Order of Volatility	RAM → Network → Processes → Disk → Logs/Backups
Golden Rule	Never work on original evidence — always work on a verified copy

40-Question MCQ Quiz — Test Yourself First!

Attempt all 40 questions below before checking the Answer Key. Difficulty is marked so you can gauge your preparation — a mix of Easy and Tricky (conceptual/twisted) questions covers the full syllabus. (Read it Carefully and Attempt)

- Q1.** What is the first step in the Six-Step Computer Forensics Process? *[Easy]*
- A) Preservation
 - B) Identification
 - C) Collection
 - D) Reporting
- Q2.** Which law in India primarily governs cybercrime and digital evidence? *[Easy]*
- A) Companies Act, 2013
 - B) Indian Penal Code only
 - C) Information Technology Act, 2000
 - D) Consumer Protection Act
- Q3.** Deleting a file from a hard disk usually means: *[Easy]*
- A) The data is instantly and permanently erased
 - B) Only the file's index/pointer is removed; data remains until overwritten
 - C) The disk is automatically reformatted
 - D) The file is moved to a hidden partition
- Q4.** In the MAC timestamp model, what does the 'A' stand for? *[Easy]*
- A) Archived
 - B) Assigned
 - C) Accessed
 - D) Authenticated
- Q5.** Which file system uses a Master File Table (MFT) instead of a File Allocation Table? *[Easy]*
- A) FAT16
 - B) FAT32
 - C) NTFS
 - D) exFAT
- Q6.** A suspect renames a malicious 'virus.exe' file to 'photo.jpg' to hide it. Which forensic concept will still reveal its true type? *[Tricky]*
- A) File extension
 - B) File signature (magic number)
 - C) File size
 - D) Folder location
- Q7.** What is 'slack space' in a file system? *[Tricky]*
- A) Space reserved for the operating system only
 - B) Leftover unused space at the end of the last allocated cluster of a file
 - C) A backup partition
 - D) Space used exclusively by malware
- Q8.** What does 'Chain of Custody' primarily establish? *[Easy]*
- A) The technical specifications of the evidence device

- B) An unbroken, documented trail of who handled evidence and when
- C) The cost of the investigation
- D) The brand of forensic software used

Q9. Which of these is generally the FIRST thing to do in First Response at an incident scene? *[Easy]*

- A) Immediately shut down all systems
- B) Begin deep analysis of files
- C) Preserve the scene and evidence without altering it
- D) Reformat affected drives

Q10. Two hexadecimal digits together represent: *[Tricky]*

- A) 1 bit
- B) 4 bits (a nibble)
- C) 8 bits (1 byte)
- D) 16 bits

Q11. What is the hexadecimal file signature of a JPG image? *[Tricky]*

- A) 25 50 44 46
- B) FF D8 FF
- C) 50 4B 03 04
- D) 4D 5A

Q12. Why is a 'hash before imaging' calculated? *[Easy]*

- A) To compress the evidence file
- B) To record the digital fingerprint of the ORIGINAL evidence before any copy is made
- C) To encrypt the original disk
- D) To speed up the imaging process

Q13. If the hash of the original evidence does NOT match the hash of the forensic image, what should the investigator do? *[Tricky]*

- A) Ignore it, since minor mismatches are normal
- B) Immediately investigate the cause — never ignore a hash mismatch
- C) Delete the image and start a new unrelated case
- D) Use only the original from then on

Q14. Which Windows Registry hive stores user account names and their hashed passwords? *[Tricky]*

- A) SOFTWARE
- B) SYSTEM
- C) SAM
- D) SECURITY

Q15. Which Registry hive would most likely show a user's recently opened documents and USB device history? *[Tricky]*

- A) SAM
- B) NTUSER.DAT
- C) SYSTEM
- D) SECURITY

Q16. What is the main advantage of Physical (Bit-by-Bit) Imaging over Logical Imaging? *[Easy]*

- A) It is always faster
- B) It only copies visible files, saving space
- C) It captures every bit including deleted files and slack space
- D) It requires no hash verification

- Q17.** What is the purpose of a write blocker during forensic imaging? *[Easy]*
- A) To speed up the copying process
 - B) To allow reading evidence while preventing any data from being written to it
 - C) To encrypt the evidence automatically
 - D) To compress the disk image
- Q18.** Imaging a suspect's drive WITHOUT a write blocker most likely results in: *[Tricky]*
- A) Faster investigation with no downside
 - B) Possible accidental modification of the evidence (e.g., updated timestamps), risking rejection in court
 - C) Automatic encryption of the drive
 - D) No effect at all on forensic value
- Q19.** Which forensic image format is created using the classic Linux 'dd' command? *[Easy]*
- A) E01
 - B) AFF
 - C) RAW (DD)
 - D) ISO
- Q20.** Which of the following BEST distinguishes Encoding from Encryption? *[Tricky]*
- A) Encoding needs a key to reverse; Encryption does not
 - B) Encoding is for format compatibility and needs no key to reverse; Encryption is for confidentiality and needs a key
 - C) They are exactly the same process
 - D) Encoding is one-way; Encryption is two-way
- Q21.** Which of these best describes Hashing? *[Easy]*
- A) A reversible process used to hide the content of data
 - B) A one-way function producing a fixed-length fingerprint used to check data integrity
 - C) A method to compress large files
 - D) A method of physically destroying data
- Q22.** What does a 'hash collision' mean? *[Tricky]*
- A) A hash calculation has failed completely
 - B) Two different pieces of data produce the same hash value
 - C) The hash algorithm has been deleted
 - D) Data has been successfully decrypted
- Q23.** 'Bit Rot' refers to: *[Tricky]*
- A) Malware that specifically targets binary files
 - B) The slow, natural degradation of stored data over time due to physical media decay
 - C) A type of ransomware attack
 - D) An encryption failure caused by wrong keys
- Q24.** What is the main difference between a capture filter and a display filter in tools like Wireshark? *[Tricky]*
- A) They are identical in function
 - B) A capture filter decides what gets recorded before capturing; a display filter only changes what is shown from already-captured data
 - C) A display filter deletes packets permanently
 - D) A capture filter only works on Windows
- Q25.** What does 'promiscuous mode' allow a network interface to do? *[Easy]*
- A) Encrypt all outgoing packets

- B) Capture all traffic on the network segment, not just packets addressed to it
- C) Block unauthorized users
- D) Automatically repair corrupted packets

Q26. Which time standard should investigators convert all logs to, when comparing evidence from devices in different time zones? *[Easy]*

- A) Local time of the investigator
- B) UTC (Coordinated Universal Time)
- C) Epoch time only
- D) Server time of the attacker

Q27. Epoch time is measured as: *[Tricky]*

- A) Seconds since 1 January 2000
- B) Seconds since the device was manufactured
- C) Seconds since 1 January 1970 (UTC)
- D) Minutes since the last reboot

Q28. An investigator sees Microsoft Word suddenly spawning a PowerShell process. What forensic red flag does this represent? *[Tricky]*

- A) Normal software update behaviour
- B) An unusual parent-child process relationship, often a sign of malware/attack
- C) A hardware failure
- D) A scheduled backup task

Q29. Why do malware authors often use multiple threads within a single process? *[Tricky]*

- A) To make the malware run slower and stealthier only
- B) To perform multiple malicious tasks at once using fewer resources, making detection harder
- C) Threads have no real advantage for malware
- D) To automatically alert antivirus software

Q30. Which Linux file records commands typed by a user in the terminal? *[Easy]*

- A) auth.log
- B) syslog
- C) bash_history
- D) /proc

Q31. Which Windows Event category would help detect a brute-force login attack? *[Tricky]*

- A) Process Creation
- B) Logon/Logoff (including failed attempts)
- C) Account Changes only
- D) Privilege Escalation only

Q32. What is the 'Golden Rule' of forensic evidence handling? *[Easy]*

- A) Always work directly on the original evidence for accuracy
- B) Never work on the original evidence — always analyse a verified copy
- C) Only use free/open-source tools
- D) Always destroy evidence after the case closes

Q33. In Live System Forensics, what does the 'Order of Volatility' principle recommend collecting FIRST? *[Tricky]*

- A) Archived backups
- B) Hard disk files
- C) RAM / memory contents

D) Printed logs

Q34. Why are Faraday bags used in Mobile Forensics? *[Easy]*

- A) To charge the phone faster
- B) To block wireless signals and prevent remote wipe or data alteration
- C) To improve the phone's camera quality
- D) To decrypt the phone automatically

Q35. What is the primary purpose of Accreditation Standards (e.g., ISO/IEC 17025) in forensic labs? *[Tricky]*

- A) To reduce lab costs
- B) To provide a benchmark proving the lab/process is competent and trustworthy, strengthening evidence credibility in court
- C) To make investigations faster only
- D) To replace the need for chain of custody

Q36. Which tool is primarily used to CREATE and PREVIEW forensic disk images (not deep analysis)? *[Tricky]*

- A) Full FTK (Forensic Toolkit)
- B) FTK Imager
- C) Sysinternals Suite
- D) Wireshark

Q37. What does a Bootloader do during system startup? *[Easy]*

- A) Displays the login screen
- B) Loads the operating system kernel into memory after BIOS/UEFI hands over control
- C) Performs the Power-On Self-Test (POST)
- D) Manages user permissions after login

Q38. A company employee uses office internet for personal shopping during work hours, against policy but without breaking any law. This is best classified as: *[Tricky]*

- A) A computer crime with criminal liability
- B) An unauthorized activity (policy violation), not necessarily a computer crime
- C) A cyber-terrorism offence
- D) Identity theft

Q39. Why is Logical Imaging sometimes insufficient for a thorough forensic investigation? *[Tricky]*

- A) It takes far too long to complete
- B) It does not capture deleted files, slack space, or unallocated space that Physical Imaging would capture
- C) It cannot be hashed for verification
- D) It only works on Linux systems

Q40. What is the key legal risk if an investigator collects far more personal data than necessary for a case (e.g., unrelated private photos)? *[Tricky]*

- A) No risk — more data is always better for a case
- B) It may violate the subject's privacy rights and could make the evidence inadmissible
- C) It automatically speeds up the trial
- D) It has no effect on the case outcome

Answer Key & Detailed Explanations

Use this section to check your answers and understand *WHY* each answer is correct — this is where the real learning happens.

Q1. What is the first step in the Six-Step Computer Forensics Process?

Correct Answer: B) Identification [Easy]

Explanation: The process always begins with Identification — recognising that an incident occurred and identifying possible sources of evidence — before anything is preserved or collected.

Q2. Which law in India primarily governs cybercrime and digital evidence?

Correct Answer: C) Information Technology Act, 2000 [Easy]

Explanation: The IT Act, 2000 (amended 2008) is India's primary cyber law, covering hacking, data protection, digital signatures, and cybercrime punishments.

Q3. Deleting a file from a hard disk usually means:

Correct Answer: B) Only the file's index/pointer is removed; data remains until overwritten [Easy]

Explanation: Deletion typically only removes the FAT/MFT entry pointing to the data. The actual bits stay in unallocated space until new data overwrites them — the basis of data recovery.

Q4. In the MAC timestamp model, what does the 'A' stand for?

Correct Answer: C) Accessed [Easy]

Explanation: MAC = Modified, Accessed, Created. 'Accessed' records when a file was last opened or read, even without any change to its content.

Q5. Which file system uses a Master File Table (MFT) instead of a File Allocation Table?

Correct Answer: C) NTFS [Easy]

Explanation: NTFS (New Technology File System) uses the MFT, a detailed database giving each file its own record with metadata and permissions — unlike FAT32's simpler allocation table.

Q6. A suspect renames a malicious 'virus.exe' file to 'photo.jpg' to hide it. Which forensic concept will still reveal its true type?

Correct Answer: B) File signature (magic number) [Tricky]

Explanation: The file signature (the first few bytes, e.g., 4D 5A for EXE) cannot be changed simply by renaming the extension. Forensic tools check the signature, exposing the disguise.

Q7. What is 'slack space' in a file system?

Correct Answer: B) Leftover unused space at the end of the last allocated cluster of a file [Tricky]

Explanation: Because file sizes rarely fill a cluster exactly, the remaining unused portion (slack space) can retain fragments of previously deleted data — valuable to investigators.

Q8. What does 'Chain of Custody' primarily establish?

Correct Answer: B) An unbroken, documented trail of who handled evidence and when [Easy]

Explanation: Chain of custody is the documented history proving evidence was properly controlled and not tampered with — essential for admissibility in court.

Q9. Which of these is generally the FIRST thing to do in First Response at an incident scene?

Correct Answer: C) Preserve the scene and evidence without altering it [Easy]

Explanation: First Response focuses on preserving the scene and evidence exactly as found — analysis and any drastic action come only after proper preservation and documentation.

Q10. Two hexadecimal digits together represent:

Correct Answer: C) 8 bits (1 byte) [Tricky]

Explanation: Each hex digit represents 4 bits (a nibble); two hex digits together represent 8 bits, which equals exactly one byte — the reason hex is convenient for viewing raw data.

Q11. What is the hexadecimal file signature of a JPG image?

Correct Answer: B) FF D8 FF [Tricky]

Explanation: FF D8 FF is the standard starting signature of JPEG images. 25 50 44 46 (%PDF) is for PDFs, 50 4B 03 04 is ZIP/DOCX, and 4D 5A (MZ) is for EXE files.

Q12. Why is a 'hash before imaging' calculated?

Correct Answer: B) To record the digital fingerprint of the ORIGINAL evidence before any copy is made [Easy]

Explanation: The hash before imaging captures the exact fingerprint of the original evidence so it can later be compared against the image's hash to prove the copy is identical.

Q13. If the hash of the original evidence does NOT match the hash of the forensic image, what should the investigator do?

Correct Answer: B) Immediately investigate the cause — never ignore a hash mismatch [Tricky]

Explanation: A hash mismatch means the copy is not identical to the original. Ignoring it makes the evidence legally worthless; the cause must always be investigated.

Q14. Which Windows Registry hive stores user account names and their hashed passwords?

Correct Answer: C) SAM [Tricky]

Explanation: SAM (Security Account Manager) stores local user accounts and their hashed passwords. SYSTEM holds hardware/driver config, SOFTWARE holds installed apps, and SECURITY holds security policy.

Q15. Which Registry hive would most likely show a user's recently opened documents and USB device history?

Correct Answer: B) NTUSER.DAT [Tricky]

Explanation: NTUSER.DAT is the per-user profile hive, storing personal settings such as recently opened files, USB history, and desktop preferences.

Q16. What is the main advantage of Physical (Bit-by-Bit) Imaging over Logical Imaging?

Correct Answer: C) It captures every bit including deleted files and slack space [Easy]

Explanation: Physical imaging copies the entire disk bit-for-bit, including deleted files, slack space, and unallocated space — unlike logical imaging, which only copies files visible to the OS.

Q17. What is the purpose of a write blocker during forensic imaging?

Correct Answer: B) To allow reading evidence while preventing any data from being written to it [Easy]

Explanation: A write blocker (hardware or software) lets an investigator read a device fully while physically/logically blocking any write commands, preserving the original evidence unchanged.

Q18. Imaging a suspect's drive WITHOUT a write blocker most likely results in:

Correct Answer: B) Possible accidental modification of the evidence (e.g., updated timestamps), risking rejection in court [Tricky]

Explanation: Even simply connecting a drive without a write blocker can trigger writes like updated access timestamps, contaminating evidence and threatening its admissibility.

Q19. Which forensic image format is created using the classic Linux 'dd' command?

Correct Answer: C) RAW (DD) [Easy]

Explanation: RAW/DD format is a plain bit-for-bit copy, historically created with the Unix/Linux 'dd' command, and is universally supported by forensic tools.

Q20. Which of the following BEST distinguishes Encoding from Encryption?

Correct Answer: B) Encoding is for format compatibility and needs no key to reverse; Encryption is for confidentiality and needs a key [Tricky]

Explanation: Encoding (e.g., Base64) simply changes data format for compatibility and can be reversed by anyone without a key. Encryption intentionally hides data and needs the correct key to reverse.

Q21. Which of these best describes Hashing?

Correct Answer: B) A one-way function producing a fixed-length fingerprint used to check data integrity [Easy]

Explanation: Hashing produces a fixed-length 'fingerprint' and is one-way — you cannot get the original data back from a hash. It is used to verify data integrity, not for confidentiality.

Q22. What does a 'hash collision' mean?

Correct Answer: B) Two different pieces of data produce the same hash value [Tricky]

Explanation: A hash collision occurs when two different inputs accidentally (or deliberately) generate the identical hash output — this is a known weakness of MD5, which is why SHA-256 is now preferred.

Q23. 'Bit Rot' refers to:

Correct Answer: B) The slow, natural degradation of stored data over time due to physical media decay [Tricky]

Explanation: Bit rot is the gradual, natural corruption of stored data (e.g., due to magnetic decay or cosmic radiation) — not caused by any human action or malware.

Q24. What is the main difference between a capture filter and a display filter in tools like Wireshark?

Correct Answer: B) A capture filter decides what gets recorded before capturing; a display filter only changes what is shown from already-captured data [Tricky]

Explanation: Capture filters are applied BEFORE capture and reduce what's actually recorded. Display filters are applied AFTER capture and only affect what you currently see — nothing is deleted.

Q25. What does 'promiscuous mode' allow a network interface to do?

Correct Answer: B) Capture all traffic on the network segment, not just packets addressed to it [Easy]

Explanation: In promiscuous mode, a network card captures every packet passing through the segment, regardless of the destination address — essential for full network monitoring.

Q26. Which time standard should investigators convert all logs to, when comparing evidence from devices in different time zones?

Correct Answer: B) UTC (Coordinated Universal Time) [Easy]

Explanation: UTC provides one consistent global time reference. Converting all logs to UTC avoids confusion and timeline errors caused by different local time zones.

Q27. Epoch time is measured as:

Correct Answer: C) Seconds since 1 January 1970 (UTC) [Tricky]

Explanation: Epoch (Unix) time counts the number of seconds that have elapsed since 1 January 1970, 00:00:00 UTC — a standard internal time format used by most computer systems.

Q28. An investigator sees Microsoft Word suddenly spawning a PowerShell process. What forensic red flag does this represent?

Correct Answer: B) An unusual parent-child process relationship, often a sign of malware/attack [Tricky]

Explanation: Legitimate applications rarely spawn unrelated system tools like PowerShell. Such unusual parent-child process relationships are a classic indicator of malicious activity.

Q29. Why do malware authors often use multiple threads within a single process?

Correct Answer: B) To perform multiple malicious tasks at once using fewer resources, making detection harder [Tricky]

Explanation: Threads allow a single process to do several things simultaneously (e.g., stealing data while hiding itself) efficiently, which can make the malicious activity harder to detect.

Q30. Which Linux file records commands typed by a user in the terminal?

Correct Answer: C) bash_history [Easy]

Explanation: bash_history stores a record of commands a user has typed, though attackers frequently try to clear or tamper with this file to hide their actions.

Q31. Which Windows Event category would help detect a brute-force login attack?

Correct Answer: B) Logon/Logoff (including failed attempts) [Tricky]

Explanation: Logon/Logoff events, especially repeated failed login attempts, are the primary indicator investigators check to detect brute-force password attacks.

Q32. What is the 'Golden Rule' of forensic evidence handling?

Correct Answer: B) Never work on the original evidence — always analyse a verified copy [Easy]

Explanation: The foundational principle of digital forensics is to preserve the original evidence untouched and perform all analysis on a properly verified (hashed) copy.

Q33. In Live System Forensics, what does the 'Order of Volatility' principle recommend collecting FIRST?

Correct Answer: C) RAM / memory contents [Tricky]

Explanation: The Order of Volatility says the most volatile data (which disappears fastest) should be collected first — RAM contents are lost the instant power is cut, so they come before disk data or backups.

Q34. Why are Faraday bags used in Mobile Forensics?

Correct Answer: B) To block wireless signals and prevent remote wipe or data alteration [Easy]

Explanation: Faraday bags block all incoming/outgoing wireless signals, preventing a suspect (or remote attacker) from triggering a remote wipe or altering evidence on a seized mobile device.

Q35. What is the primary purpose of Accreditation Standards (e.g., ISO/IEC 17025) in forensic labs?

Correct Answer: B) To provide a benchmark proving the lab/process is competent and trustworthy, strengthening evidence credibility in court [Tricky]

Explanation: Accreditation standards prove a forensic lab follows recognised, competent processes, which increases the credibility and court-acceptance of evidence it produces.

Q36. Which tool is primarily used to CREATE and PREVIEW forensic disk images (not deep analysis)?

Correct Answer: B) FTK Imager [Tricky]

Explanation: FTK Imager is specifically for creating and previewing forensic images without altering evidence. The full FTK suite is used afterward for in-depth analysis of that evidence.

Q37. What does a Bootloader do during system startup?

Correct Answer: B) Loads the operating system kernel into memory after BIOS/UEFI hands over control [Easy]

Explanation: After BIOS/UEFI completes POST and locates the boot device, the bootloader (e.g., GRUB, Windows Boot Manager) takes over and loads the OS kernel into memory.

Q38. A company employee uses office internet for personal shopping during work hours, against policy but without breaking any law. This is best classified as:

Correct Answer: B) An unauthorized activity (policy violation), not necessarily a computer crime [Tricky]

Explanation: This behaviour violates company policy (an unauthorized activity) but does not break any specific law, so it is not classified as a computer crime — only internal action would typically follow.

Q39. Why is Logical Imaging sometimes insufficient for a thorough forensic investigation?

Correct Answer: B) It does not capture deleted files, slack space, or unallocated space that Physical Imaging would capture [Tricky]

Explanation: Logical imaging only copies files/folders visible to the OS, missing deleted data, slack space, and unallocated space — which is why Physical (bit-by-bit) imaging is preferred for complete investigations.

Q40. What is the key legal risk if an investigator collects far more personal data than necessary for a case (e.g., unrelated private photos)?

Correct Answer: B) It may violate the subject's privacy rights and could make the evidence inadmissible

[Tricky]

Explanation: Collecting data beyond what is relevant and legally authorised can violate privacy rights, and courts may reject evidence gathered through such overreach.

How Did You Score?

Score	What It Means
34 - 40	Excellent! You have strong command of the whole syllabus.
26 - 33	Good — revisit the topics behind the questions you missed.
18 - 25	Fair — re-read the Units where you struggled, then retake the quiz.
Below 18	Go back through the notes unit by unit before attempting again — focus on the ★ IMPORTANT boxes first.